BIT 4105: ADVANCED DATA STRUCTURES AND ALGORITHMS

CAT 2

QUESTION ONE

a) Describe the concept of the queue ADT and write algorithms to implement the initialization, Enqueue and Dequeue operations on a queue. [10 marks]

b) Use the following data; **33, 10, 13, 12, 45** Using a hash key of 8 enter the data in an hash table [10 marks]

QUESTION TWO

a) Showing how a Huffman tree can be used for data compression using Huffman Coding [5 marks]

b) Write an algorithms that performs the following operations on a graph;

(i) Breadth first search [5 marks]

(ii) Depth first search [5 marks]

c) By the use of a relevant example, demonstrate how a heap tree can be used for sorting a set of data. [5 marks]

QUESTION THREE

a) For each of the following two algorithms, give its time complexity in Big-oh notation [8 marks]

(i) Algorithm Algol (A)

Input: An array A storing n>=1 integers

Output: The sum of the elements in A

```
s←A[0]
for i←1 to n-1 do
  s←s + A[i]
  return s
```

(ii) Algorithm Algol (A)

Input: An array A storing n>=1 integers

Output:The sum of the prefix sums in A

```
s←0
{
for i←1 to n-1 do
      s←s + A[0]
for j←1 to i do
s←s + A[j]
}
 return s
```

b) Write pseudo code for a binary search tree. Assume the array is already sorted. [5 Marks]

c) Using a flow-chart, represent the algorithm for a pop operation in a stack [3 marks]

d) Giving an example describe what a recursive function is. [4 marks]

Question Four

a) Explain the difference between Kruskal's Algorithm and Prim's Algorithm. [4 marks]

b) Describe depth-First Search in algorithms. [4 marks]

c) Describe any four application areas of graph theory. [8 marks]

d) Define the following terms as used in analysis of algorithms. [4 marks]

   i. Euler circuit.

   ii. Complete Graph**.**

Question Five

a) Describe the structure Greedy Algorithm. [4 marks]

b) Using example, write an algorithm that illustrates the following time complexities

   I. O(n)

   II. $O(n^2)$ [6 marks]

c) Discuss two main areas of applications for Binary First Search. [6 marks]

d) Discuss the searching problem and state any two searching approaches [4 marks]

Question Six

a) Using a real life situation explain how **Activity Selection** Problem can be useful, use a pseudo code to explain your answer. [4 marks]

b) Explain the difference between forward and back edge in graph algorithm.[4 marks]

c) Differentiate between a recursive algorithm and recursive procedure. [2 marks]

d) Design a car renting system, including reserving a car, checking in and checking out. Consider all the cases: reserve a car, then check out successfully; reserve a car, but the car is sold out before you check out. [10 marks]